

**D13: Shea Management Plan
(English)**

1. Introduction

When clearing savannas for agricultural production purposes, farmers select and conserve on their agricultural land the plants that are promising or close to the productive age of certain utilitarian woody species, including shea (*Vitellaria paradoxa* Gaertn .f.), which thus become an integral part of agricultural systems. However, throughout their range, shea stands are experiencing increasing degradation due to the disappearance of fallows, the modernization of agriculture, the cutting of trees for the production of firewood and charcoal, the establishment of other crops as well as drought.

In the north of its distribution area, due to drought, we can see the aging of shea parklands, the mortality of certain trees and the lack of regeneration. In areas of land saturation such as the outskirts of large urban areas, permanent cultivation maintains pre-existing trees which benefit from privileged conditions, in particular through weeding and soil fertilization operations. On the other hand it is detrimental to the future of the population through the repeated weeding of the fields which leaves no possibility of development for young shea seedlings. Even where climatic conditions are more favorable, parklands also age and are not replaced by sufficient regeneration, especially since the use of the plow (and even more that of the tractor!) often results in the uprooting of young shoots. This situation is exacerbated by the fact that the shea parklands are considered by local communities as common goods making their management difficult.

Even if nowadays some settlements seem to develop in a context of low anthropic and animal pressure, it is not excluded that in the near future conflicts will arise between the members of the community to appropriate the land and compromise the existence of the stand because, with the establishment of crops, some trees will be systematically felled.

A main source of vegetable fat, shea butter has always been used in the Sudano-Sahelian zone of Africa for domestic consumption. Most of the harvested fruit - 80 to 90% depending on the source and the year - is kept for home consumption or for sale in local markets. The rest, or 10 to 20%, is marketed, either in the form of nuts or in the form of butter. Most of the exported butter is used in the food industry, mainly in chocolate and confectionery. Shea is currently attracting growing interest since the European Union's decision to allow 5% vegetable fat in the manufacture of chocolate as a substitute for cocoa butter. The market for artisanally produced shea butter (5%) by women and intended for export supplies cosmetic companies. These are niche markets (often organic and / or fair trade) for which demand in terms of volume is relatively low but supplies are regular and constant.

Women are the main players in the shea sector, which thus provides them with income, greater financial autonomy, and helps reduce the social divide between men and women. The exploitation of the fruits of the shea tree therefore constitutes an opportunity (especially with the possibilities of exporting) to reduce rural poverty and offers a development path for women who make up the poorest fringe of the rural population. However, the overexploitation of the potential is currently being done without any concern for the preservation and renewal of the resource which seems inexhaustible given the longevity of the shea tree. Poor populations, who overexploit their immediate environment to fight poverty, are very vulnerable to climate change, especially extreme climatic situations such as droughts and floods.

Producers of shea-based products are aware of the degradation of the resource. However, the traditional passivity of tree management practices in a demographic context that has dramatically changed and the lack of scientific information limit the potential for development and above all for the sustainable management of shea parklands. Management techniques (maintenance of producing trees and regeneration) of stands exist but they need to be improved and are not always mastered by producer organizations who have difficulty in applying them.

To compensate for the decrease in the resource and production of shea nuts, for domestic consumption and for export, it is necessary on the one hand to strengthen the capacities of women producers in action research in the field of management of shea stands as an agroforestry system, and on the other hand to enable them to be able to conduct the silvicultural operations necessary to increase tree production while taking into account the climatic zones of the country.

2. Shea parklands

2.1. What is shea ?

Shea (*Vitellaria paradoxa* Gaertn .f.) or “butter tree” is a Sapotaceae typical of the Sudano-Sahelian zone between 600-1500 mm isohyets. It is the dominant tree species in many agroforestry parklands in Sub-Saharan Africa. Its range stretches from Senegal in the west to Uganda in the east over a strip approximately 5,000 km long and 400 to 750 km wide. There are two subspecies depending on the fatty acid composition of the kernels: *Vitellaria paradoxa* subsp. *paradoxa* which is present in West and Central Africa, and *Vitellaria paradoxa* subsp. *nilotica* found in East Africa. Shea nut provides butter, a quality vegetable fat that is used for cooking, cosmetics and pharmacopoeia.

2.2. Importance of shea

Importance of shea for women

Women are the main players in the shea sector (90% of stakeholders in Burkina Faso) which thus provides them with income and greater financial autonomy. The income generated by the sale of butter and kernels allows women to meet the daily needs of the family (children's health and education costs) and even sometimes the needs of the community during social events. They thus contribute to the socioeconomic development of their village. A study by the Global Shea Alliance (GSA) shows that the shea industry has nearly doubled over the past decade and currently provides jobs for 16 million rural women in 21 African countries.

Importance of shea in national economies

In addition to its many local uses (cooking, pharmacopoeia, soap, etc.), shea is one of the main export products of several African countries (shea products are the 3rd export earner in Benin and the 4th in Burkina Faso). Shea butter is an important component of food (margarines, chocolate, etc.), cosmetics and pharmaceuticals around the world. The exploitation of nuts therefore is an opportunity (especially with export possibilities for the cosmetics and food industry) to reduce rural poverty and offers a development path for countries.

2.3. Evolution trends of shea parklands

Global negative trends

Shea parklands are experiencing increasing degradation due to climatic factors (climate change, drought, high winds), health (animal and plant parasites), anthropogenic pressure (demographic and land pressure, urbanization), their poor management (excessive cutting, straying of animals / overgrazing) and unsuitable cultivation practices (clearing, agricultural mechanization, lack of regeneration and planting). Parklands are ageing and not being replaced by sufficient regeneration.

Given the pressure on land due to population growth and the scarcity of arable land, we are witnessing the decrease (in space and time) or even the disappearance of fallows resulting in insufficient regeneration of shea even where the climatic conditions are favorable. Traditionally, after a cultivation phase, cultivated plots were left fallow for a period which could exceed twenty years, which allowed regeneration of the woody stand. Today, the short duration of fallow (2-3 years) when it still exists, is responsible for the reduction in natural regeneration because young plants are very small and therefore often not protected, especially by non-owner farmers.

New intensive crops have appeared recently and require large spaces. Thus "agribusiness" and cotton, motorized crops, have a devastating effect on the shea resource by requiring the trees to be pulled up.

The degradation of parklands is reflected not only by a decrease in the density of shea plants but also by a genetic impoverishment and dominance of "bad" shea, that is to say of shea butter whose almonds give poor butter. quality. Regeneration can be important but consist of poor quality trees, with the nuts from the best performing trees generally being collected entirely.

In addition, the exploitation of the potential is currently done without any concern for the preservation and renewal of the resource and the collection of immature fruits prevents germination and has a negative effect on the quality of the butter. Fruit sold in town and on the side of the road is wasted and cannot contribute to regeneration.

Recent positive development in some cases

However, due to growing awareness among local populations, the presence of women's organizations involved in parkland management and the existence of local shea parkland management conventions, the spontaneous protection of young plants during agricultural work and against fire and cattle is becoming more common. Thus the number of young plants spared in fields increases.

Following interest in possibilities for export of shea kernels and butter, there is also a growing demand for grafted or supposedly improved shea plants as well as training in grafting and vegetative propagation techniques.

3. Obstacles to improved management of shea parklands

3.1. The traditional lack of planting practice

Shea trees are always the result of natural regeneration through seed germination on the ground or vegetative propagation of shoots from stump regrowth. Rural African populations are not used to plant local species, especially as shea takes many years to produce. As the sayings "if you plant a shea tree, you will not reap its fruits" or even "a farmer will die before seeing the fruits of a young shea tree" indicate, whoever plants it will not benefit from its production. From this were born legends related to the planting of shea due to its long gestation period such as "whoever plants a shea tree will die when the tree begins to bloom".

3.2. Application of legal texts

In most shea producing countries, the laws prohibit cutting shea for fuelwood or charcoal production. However, illegal cutting is common.

The ban on cutting concerns mature trees and not regeneration and opposes the adoption of techniques for the rejuvenation of shea populations: the owners of fields do not cut old trees that are not very productive but focus on the youngest not yet in age of production. If he judges the stand of his field too dense, the grower will pull up the non-productive young people not affected by the texts rather than cutting the older trees.

3.3. Tenure issues

In rural Africa, traditional customary law prevails in land matters.

Owners / non-owners

Land is accessed under the permission of the "landowner". The loan is free and depending on the location the tenant donates part of his cereal production but he cannot always enjoy the fruits of shea trees because the tree itself remains the property of the land owner. Only owners can plant shea. Non-owners cannot plant or cut trees. When they cannot take advantage of shea fruits and kernels and tree shade reduces crop yields, non-owners do not favor the practice of fallow, nor any other practice such as the protection of young shoots allowing the renewal of the woody stand. They even go so far as to destroy agroforestry parklands.

Men/women

As a rule, in rural areas land and trees belong to men (with a few exceptions) while shea kernels are traditionally harvested by women. The customary law of rural women is at issue. While being the main players in the shea sector, both from the point of view of harvesting, processing and marketing, they are often kept apart from the management of land and natural resources. And are not allowed to take initiatives even though they are directly affected by poor practices and natural attacks to which the species is subjected. On the other hand, there is often an opposition between the densification of the shea population and ensuring the necessary space for traditional crops (shading, spacing to allow the passage of the plow) which can lead to conflicts.

Local strategies must be developed to resolve land issues, for example getting customary chiefs to agree to allocate portions of land to groups of women producers who will be able to manage them freely and subsequently establish title deeds on the shea parklands which have been allocated to them. They will thus be able to improve the shea resource through new techniques.

Land owners/migrants

Traditional customary law gives migrants very little access to shea products. They often do not have permission to plant or graft shea trees or risk triggering conflicts with landowners.

The actions of agribusinessmen in particular as well as motorized cotton cultivation have a devastating effect on the shea resource. There are difficulties and conflicts with landowners or land chiefs when third parties (agro businessmen / civil servants) buy and establish land titles with people who do not have long term landholding rights.

4. Objectives of the document

The sustainable management of shea stands relies above all on actors' capacities to understand and influence the factors, challenges and opportunities that have a bearing in parkland management, and to apply strategies and techniques for improved management of stands and trees. This document presents a training strategy for women's groups and farmers in shea communities for improved management of shea parklands.

The objective of the training is to increase access of women collectors and farmers to shea products in a sustainable manner by offering them technical means to conserve and restore shea parklands. They will thus be better able to fight poverty and adapt to climate change. Their capacities in the management of shea stands as agroforestry systems will be strengthened and they will be able to conduct the silvicultural operations necessary to compensate for the decline in the resource and shea nut production for domestic consumption and export. The frequency of destructive practices (bush fires, uncontrolled cutting, harvesting of immature fruits, night harvest, etc.) will be reduced. These trainings, also intended to strengthen the organizational, leadership and technical capacities of women's groups, will enable them to be recognized and valued at the level of their village

and thus to intervene in decisions affecting the management of parklands and give added value to shea butter. Finally, these trainings will allow better coordination and cooperation of the various village actors for the protection and restoration of parklands in order to bypass the obstacles listed above, in particular the risk of land conflicts.

The document will allow the Serious Shea consortium to design action research and training activities for sustainable parkland management and to mentor women's groups and farmers during these trainings.

5. How to sustainably manage shea parklands: best practices for managing shea parklands

It is important to take into account that we can regenerate / protect / manage local species, especially shea, just like exotic / commercial species that are traditionally cultivated (planting, pruning, grafting, etc.).

To achieve an optimized density of shea trees comprising trees of various ages that allows for renewal of the stand, it is preferable to regenerate shea trees in agroforestry parklands itself either by protecting the young shea trees resulting from natural regeneration (RNA), either by restoring / enriching the existing stand with improved shea: grafting wildlings, planting, direct sowing from material whose superior characteristics are documented.

Shea tree management practices currently exist to mitigate infestations by parasitic phanerogams that reduce tree production and even longevity, to invigorate low-productivity trees by pruning crowns, and to regulate fruit production through girdling the main branches or boles.

6.1. Steps for the implementation of training in sustainable management of shea parklands

Six successive main steps are necessary for the implementation of training in the sustainable management of shea parklands:

Step 1: Information and Awareness

Step 2: Diagnosis

Step 3: Development of local agreements / collective tools for the management of shea parklands

Step 4: Knowledge of technical options available for the sustainable management of shea parklands

Step 5: Training and dissemination of improved management techniques for shea trees and parklands

Step 6: Participatory monitoring and evaluation

6.2. Step 1. Information and Awareness

Prior to the start of activities, a communication campaign on trends and factors in the development of parklands, as well as challenges and opportunities for the sustainable management of tree populations and change of behavior will make it possible to introduce the project and allow for objectives to be met.

Objectives

- Contribute to the improvement of knowledge on shea, trends and factors in the development of parklands, as well as challenges and opportunities for the sustainable management of tree populations.
- Sensitize local people (mainly producers but also landowners) to the need to protect and renew shea populations in the fields.

- Make producers aware of best fruit collection practices (do not harvest unripe fruits)
- Present the training process and content

Constraints/solutions

Constraints	Solutions
Poorly adapted awareness-raising materials	Develop training materials adapted to realities and the degree of understanding of local populations
Choice of language	Adapt tools to groups / beneficiaries
Limited financial and human resources	Mobilize internal resources (financial and human)
Sensitization period, availability of populations	Set a consensus date that takes into account the availability of populations
Motivation of state services (agriculture, environment, etc.)	Involve state service agents in formulating messages. Help them understand that training also concerns technicians
Poor wording of messages	Clearly formulate messages based on level of understanding of populations and targeted behavior changes
Low knowledge of dynamics, customs and traditions (local sensitivities)	Know local susceptibilities and modes of organization. Inform
Lack of motivation of the populations	Involve women's organizations (butter producers) Check the adequacy between training and real needs of populations

How to proceed?

- Duration of sensitization: 1 day / site
- Involve local and customary authorities in the process in order to benefit from their endorsement and their commitment to promote technology transfer.
- Utilize/add value to existing human resources at village level
- Involve customary authorities so they give their opinion
- Involve village development councils (CVDs) in the organization throughout the process
- Use training techniques and dissemination channels accessible to local people
- Set up thematic and multidisciplinary facilitation groups

What messages to convey?

- Messages on the need to change behavior: Inappropriate practices such as bush fires, cutting, harvesting unripe fruits, night harvesting... must be abandoned. Adding value to butter and improving shea parklands make it possible to fight against cutting.
- Messages on the importance of sustainable parkland management: Shea revenues are linked to parkland management activities. Shea can be managed by horticultural techniques just like exotic species (mango, cashew).
- Messages on the importance of shea: its economic, food and nutritional, medicinal, environmental values

Who are messages for?

- Especially to the target groups / actors concerned: women, young people, grassroots community organizations living in shea production villages that supply Serious Shea
- but also to the local administrative, customary and religious authorities of these villages and municipalities, as they are also involved in some of the management aspects of parklands identified by Serious Shea
- In addition, they are also aimed at notables because they play a role in the development of villages from which they come; to state services for administrative, political and legal questions relating to the management of parklands and the shea sector; and vocational schools if we plan to train young people in the issues and solutions regarding the sustainable management of shea parklands

Dissemination techniques and channels

Dissemination techniques

- Meetings of producer groups and village assemblies
- Feedback meetings
- Demonstration plays, sketches
- Extension support: posters, adapted technical sheets
- Seminars, workshops and conferences
- Radio / television broadcasts (depending on the desired awareness scale)

Distribution channels in production areas affiliated with Serious Shea

- Proximity radios
- Theaters
- Town criers
- Institutional channels (town hall, etc.)
- Radios, television (depending on the desired awareness scale)
- Website and social networks
- Written press

6.3. Step 2. Diagnostic

A diagnostic exercise targeted at identifying stakeholder groups and the key issues underlying the protection of the shea resource and the renewal of shea parklands will help assess the importance of the sustainable management of shea parklands for populations and will complement awareness-raising.

Objectives

- Know the distribution of the population: ethnic groups, level of education, presence or absence of migrants, percentage of owners / non-owners, association practices ...
- Know the condition of shea parklands in the area
- Document recent trends and shea management practices according to land tenure (by owners, indigenous non-owners, migrants on lineage cultivated land, fallows and bush areas) and gender issues (heads of households, women, young people, etc.)
- Test the knowledge of local populations on the shea tree, shea products, condition of parklands, good and inappropriate practices
- Evaluate the degree of awareness of local populations regarding resource conservation, known and applied protection and regeneration techniques for tree populations in the area, local knowledge and know-how on sustainable management and tree regeneration in parklands in general and for shea in particular
- Know the problems surrounding the protection and renewal of shea parklands
- Target among available management techniques those that interest local populations

Constraints/solutions

Constraints	Solutions
Inadequacy of the diagnostic method / tools	Adapt methods and tools to socio-cultural and economic realities of rural populations
Choice of language	Adapt tools to groups / respondents

	Add value to local human resources
Insufficient professionalism in the collection and processing of data	Use professionals for training and supervision on diagnostics Multidisciplinary team
Availability of populations	Select a consensus date that takes into account the availability of populations
Motivation of populations	Use techniques that enhance individual skills and knowledge (focus group / interview guide, etc.) Involvement of women's organizations (butter producers)
Importance of habits and customs	Involvement of customary authorities
Wait-and-see behavior of communities (excessive expectations)	Annoncer clairement l'objectif du diagnostic et de la formation
Primacy of the community over the individual	Develop individual interview guides
Diagnostic studies without follow-up / feedback for local people	Organize data restitution / validation sessions for local groups

How to proceed ?

- Duration of surveys: 3-5 days / site
- Diagnostic planning workshop, sharing of diagnostic objectives, harmonization of concepts by involving local and customary authorities, and village development councils (CVD) at all stages in raising awareness
- Identify target groups in the local context
- Organize a preparatory meeting / pre-survey with representatives of target stakeholder groups in order to identify key questions addressing above-mentioned objectives
- Prepare the various individual and collective interview guides / questionnaires depending on target groups: shea producers, field owners, representatives of technical environmental services, project managers in the area
- Carry out data collection by enhancing human resources of villages; analyze the data and write up diagnostic results
- Present diagnosis results during village assemblies in the presence of representatives of stakeholder groups.
- Capitalize on knowledge and information acquired for designing the local convention for shea parkland management (Step no. 3) and the action plan for the management of parklands and trees (Step no. 5).

Messages to be co-created with stakeholder groups

- Diagnostic objectives
- Importance of the diagnosis for the community
- Importance of sustainable management of shea parklands for the village
- Values of shea (economic, food and nutritional, environmental, medicinal value, etc.)

Implementation leaders

Serious Shea will be responsible for designing the diagnostic. They can seek advice from research staff, professionals in participatory action-research methods, and will benefit from relying on technical officials in local government who already have in-depth knowledge of communities and production systems.

6.4. Step 3. Development of local agreements / collective management tools for shea parklands

Securing access to shea parkland sites will be based on the development of concerted management tools such as local agreements drawn up in each locality.

General objectives

- Ensure the sustainable management of shea resources by communities

Specific objectives

- Secure access to the shea resource by local groups so that investments in the protection, restoration and enhancement of shea parklands are sustainable
- Strengthen the decision-making and control powers of local authorities

Challenges/solutions

Challenges	Solutions
Establish lasting synergy between local and administrative authorities (technical services)	Ensure that each actor takes ownership of their roles and responsibilities
Contain / minimize land conflicts	Develop a preventive approach to managing conflicts related to the utilization of natural resources (shea)
Avoid untimely interventions by political and administrative authorities	Advocate at the local level for the sustainable management of parklands
Rational management of natural resources with particular emphasis on shea	Define simple and clear rules for the management of natural resources (shea)
Prevent the anarchic exploitation of forest resources by transhumants	Apply sanctions
Ensure the effective implementation of the convention	Set up a coordination unit Designate a leader Organize information and awareness sessions
Enforce the clauses of the agreement by residents	Setting up of signs Wide dissemination of the convention through local channels Translation of the agreement into local languages Apply sanctions
Avoid the non-appropriation of the agreement by local government authorities	Organize information and awareness sessions
Mobilize financial and human resources to ensure compliance with the clauses of the agreement	Have an operational local committee

How to proceed ?

Diagnose the situation (if necessary in complement to Step 2)

- Identification of cross-cutting issues and (customary and modern) management rules for shea populations that exist at each site. The assessment will be carried out through questionnaires which will be targeted for customary authorities, and local government services of agriculture, environment, livestock
- Observation of inappropriate state and / or management of resources
- Identification of strengths, weaknesses, threats and opportunities of the sector
- Identification of key players, their expectations, roles and importance

Develop a draft convention

- Identification and selection of a service provider or a resource person
- Making contact with identified key players
- Determination of roles and responsibilities of stakeholders according to the strategy
- Development of a concerted strategy governing management rules for shea tree populations in each site (who does what, how and where?)
- Address consistency of local rules with legal requirements
- Development of the draft convention

Validation and signature by the actors

- Additional visit to each site in order to present the draft convention to local populations and authorities for their review, validation and adoption.

- Discussion of the draft convention with all the stakeholders identified
- Amendment of the draft convention
- Presentation and adoption of the convention at a village assembly
- Signature of the convention document by villagers, the local administration and technical services
- Transmission to the competent authority for signature

Implementation of the convention

- Establishment of a multi-stakeholder platform
- Establishment of a management committee
- Establishment of a coordination unit
- Translation in local languages and dissemination of the agreement
- Implementation of the convention
- Monitoring and evaluation
- Organization of periodic meetings with the various actors

Working time

- Time frame for the diagnosis : 4 days/site
- Time frame for the drafting and amendment of the draft convention : 3 days/site
- Time frame for presentation and validation of the convention : 3 days/site

Messages to convey

- Importance of sustainable management
- Strong involvement of community leaders
- Necessity of strict compliance with commitments stipulated in the agreement on the roles and responsibilities of each actor
- Need for rigorous application of disciplinary standards and sanctions contained in the agreement

Actors involved

Various operators in the shea sector; Associations of women collectors; Producers; Transformers; Traders; Forest resource operators; Technical services (Water and Forests, Agriculture, Livestock); Administrative, customary and religious authorities; Territorial communities; Local communities; Research institutions and partners; NGOs; Technical and financial partners.

Actors in charge of the implementation of conventions

- Coordination Unit
- Multi-stakeholder platform
- Management committee
- Leaders of various associations and women groups
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Useful references

- Intercoopération – CORADE – FNZ & UGPPK – UPPPK & UPPC, 2012. Conventions locales : Comment s'entendre pour bien gérer les parcs agroforestiers à karité ? Burkina Faso, 18 pages
- Helvetas Swiss Intercoopération Mali. 2012. Document de capitalisation des expériences du projet Shisun: Convention locale et gestion des parcs à karité, 10 pages
- Ministère de l'Environnement et de l'Assainissement du Mali. 2011. Guide méthodologique national: Elaboration d'une convention locale de gestion des ressources naturelles (disque compact). Coopération allemande, Intercooperation, World Agroforestry Centre.

6.5. Step 4. Knowledge of the technical options available for the sustainable management of shea parklands

Management techniques (maintenance, improvement and regeneration of productive trees) of shea parklands exist but they are not always mastered by producer organizations who may find them difficult to apply.

General objective : Sustainable management of the shea resource

Specific objectives

1. Ensure the systematic regeneration of shea parklands
2. Ensure sustained fruit production in quantity and quality
3. Ensure the integrated management of agroforestry parklands and the development of the shea resource

Challenges/solutions

Challenges	Solutions
Help people to master, own and adopt good practices and techniques for sustainable management of shea	Sufficient level of awareness Choice of techniques adapted to needs and means of local people
Provide appropriate training on grafting techniques, pruning, RNA transplantation	Well mastered techniques by the implementation team
Ensure that the techniques are well mastered by local people	Effective mobilization of financial, material and human resources
Restore degraded parklands	Strong ownership of parkland maintenance and regeneration techniques by local people
Reduce the length of the shea cycle	Grafting
Provide local people with high-performance shea material	Genetic improvement
Find the financial and human resources to conduct research on shea	Create a fund for conducting research on the sustainable management of the resource and the value addition of shea

Parkland and tree management options

1. Protection of shea parklands

Depending on local needs which are to be confirmed depending on the site, several technologies can contribute to the effective management of natural resources in agroforestry parklands. These are for example:

- Protection against livestock encroachment by live fences
Référence : Fiche technique n°15 INERA « Comment choisir son espèce et sa technique pour obtenir une haie vive défensive efficace »
- Stone bunds, anti-erosion bunds, half-moons, and other techniques and practices for water and soil conservation and the improvement of agricultural production
Référence: Fiche technique n°7 INERA « Lutte anti-érosive et amélioration de la productivité de sol par l'aménagement de cordons pierreux »
- Cultural techniques that protect young shea plants
- Targeted cropping under the shade of trees (chilli, taro, yam, groundnut, beans, sweet peas, earth peas and ginger) in order to take advantage of areas under tree crowns and to prevent non-landowners from cutting shea trees due to yield decline of annual crops in the shade
- Implementation of local management agreements for shea parklands

2. Management and regeneration of shea parklands

- Assisted natural regeneration

References :

1. Samaké O., Dakouo J. M., Kalinganiré A., Bayala J., Koné B. 2011. Régénération naturelle assistée. Gestion des arbres champêtres au Sahel. Manuel technique No. 16. World Agroforestry Centre.
2. Fiche technique « La Régénération Naturelle Assistée ou Comment prendre soin de vos jeunes karités dans les champs ? » INERA.

- Grafting of wildlings

Reference : Fiche technique « Le greffage de sauvageons pour enrichir vos parcs à karité ». INERA/CNSF

- Direct sowing

Reference : Fiche technique « Le semis de graines de karité dans les buissons pour restaurer les parcs à karité » INERA / WAAPP / CNRST.

- Nursery

Reference : Fiche technique « La production de plants de karité en pépinière villageoise » INERA/CNSF

- Grafting of nursery plants

References :

1. Kambou S., Lompo D. et S. Sina. 2015. Manuel de greffage du karité (*Vitellaria paradoxa* Gaertn. f.). CNSF. Ministère de l'Environnement et des Ressources Halieuthiques, 30 pages.
2. Intercoopération – CORADE – FNZ & UGPPK – UPPPK & UPPC, 2012. La domestication du karité. Innovations techniques et savoirs paysans, 16 pages
3. Helvetas Swiss Intercoopération Mali. 2012. Document de capitalisation des expériences du projet Shisun: Domestication du karité, 10 pages
4. Chimsah, F. A., Yidana, J. A., Lovett, P.N. and Nyarko, G. 2012. (Poster) A Guide to shea grafting: A Case Research from the University For Development Studies, Tamale, Ghana. Department of Horticulture, University for Development Studies (UDS).

- Plantation of nursery-grown plants

Reference : Fiche technique « Restauration des parcs à karités par la plantation de karités produits en pépinière » INERA

- Sanitary cuts

References :

1. Samaké O., Dakouo J. M., Kalinganiré A., Bayala J., Koné B. 2011. Techniques de déparasitage et gestion du karité au champ. Manuel technique No. 15. World Agroforestry Centre.
2. Bayala, J., Ky/Dembélé C., Boussim I. J. 2012. Technique de taille sylvicole pour la lutte contre les parasites Loranthaceae et l'amélioration de la production fruitière du karité. Fiche technique INERA, 4 pages

- Rejuvenation cuts

Reference: Bayala, J., Teklehaimanot, Z. Ouedraogo S.J. 2008. La taille des arbres comme outil de rajeunissement des arbres et d'amélioration de leur production fruitière dans les parcs agroforestiers traditionnels. Fiche technique INERA, 3 pages

- Girdling

Reference : Lamien N. 2010. Fiche Technique d'accroissement de la fructification du karité (*Vitellaria paradoxa* Gaertn.) par la technique du cernage. INERA, 1 page

Research topics to explore

- Optimal density and spatial distribution of shea trees to allow both agriculture and shea development
- Quality improvement through a genetic improvement program to make shea more attractive:
 - Identification of priority traits: 1) quality and quantity of pulp for the consumption of fresh fruit, 2) butter with high fat content, percentage of stearic-oleic-stearic triglycerides for the export market of 'cocoa butter equivalent (CBE) and local consumption of butter, 3) butter with a high content of unsaturated fatty acids for the cosmetic industry, and 4) butter with high levels of unsaponifiables and phytosterols for medicinal uses, 5) adaptability to drought, 6) short gestation period, etc.
 - Evaluation of trees according to priority traits over several years and identification of "plus" trees
 - Controlled pollination and crosses of two "plus" parents in order to obtain a strong genetic gain and dissemination of this high-performance material: seeds for planting, grafts of young plants resulting from these crosses for grafting
- Design the specifications for a fresh fruit supply chain and value addition, selection of plants suitable for the production of sweet fruits, determine the optimal harvesting stage on the tree, improved fruit conservation techniques, practices for processing of fruits into jam and juice, training of women producers, etc.
- Ways to improve fruit collection, kernel pre-processing and butter processing practices.
- Specifications for the production of fuel briquettes and livestock feed pellets based on shea oilcakes

How to proceed ?

- Inform and train local groups on the various options
- Implementation and monitoring / evaluation according to the target group:
- For land-owning farmers
 - Choice of site
 - Choice of density and species to protect
 - Application of shea harvesting and management techniques (assisted natural regeneration, grafting wildlings in the field, rejuvenation cuts, pest control, tree pruning)
 - Good practices for fruit collection and kernel pre-treatment
 - Membership in the local shea parkland management convention
- For non-owner operators
 - Negotiate with the landowner authorization to apply sustainable management practices and techniques
 - Choice of site
 - Shade-adapted crops
 - Traditional mode of land transfer
- For women
 - Contribute to the identification, selection and conservation of "plus" shea trees

- Ensure good fruit collection and kernel production practices
- Foster the creation or facilitation of associations for the exchange of ideas and experiences in sustainable shea parkland management and the acquisition of plots from customary authorities

Messages to convey

- To villagers
 - Ensure best practices in development and sustainable management of parklands and natural formations (new fields, fallows and fields)
 - Need to safeguard shea trees
 - Adopt practices that ensure good quality kernels
- To research institutions, projects and NGOs
 - Provision of high-performance and well-adapted plant material
 - Provision of technical sheets on shea regeneration and management
 - Wide dissemination of new parkland management techniques

Implementation leaders

National and international research institutions, Rural development and extension services, Serious Shea, Management Committees and beneficiary farmers (men, women, youth, children), farm household heads, groups of shea collectors.

6.6. Step 5. Training and dissemination of improved tree and parkland management techniques

Selection of school plots

Constraints/solutions

Constraints	Solutions
The maximum shea tree density that the owner is ready to accept in his field has been reached (competition with crops)	Do not include the plot
Risks of conflicts within the village	Implement local conventions Involvement of customary and administrative authorities
Risk of conflict between villages benefiting or not from the project	Awareness creation can help limit the risk of conflict. See how to integrate other villages in the project

How to proceed ?

- A 100% voluntary approach is needed: the request for intervention must come from the owner of the plot himself.
- Take current trends in cultivation practices into account to avoid an opposition between the density of shea trees in the field and crops (in particular cotton with the use of the tractor does not combine well with trees). Development does not go in the direction of the return to the daba (traditional pickaxe).
- The involvement of customary and administrative authorities is necessary to limit conflicts.
- Select plots where the number of shea trees is low. Indeed, the intensification of shea production can only be sustainable if it takes into account the maximum density of shea trees that the owner is ready to accept in his field. If the stand is considered too dense, the producer will pull out young unproductive trees not concerned by legal texts rather than cutting older trees.

Implementation leaders

- Research institutes or government technicians in the environment, agriculture
- Serious Shea facilitators
- Customary and administrative authorities
- Landowners

Selection of regeneration and management techniques to use

Constraints/solutions

Constraints	Solutions
Technique recently developed by researchers	Validate the technique before transfer to farmers
The technique requires the use of equipment or chemicals deemed too expensive by producers who will not be able to use them after the end of the project.	See if the cost can be reduced by using locally available material
Results are not immediately visible nor convincing enough in the eyes of local people	Improve the technique before dissemination
The ratio of observed result / investment of time or effort is too low in the eyes of producers.	Do not propose the technique
Results / techniques do not correspond to the needs of the populations	Do not propose the technique

How to proceed ?

- Identify the real needs of local people through the diagnosis
- Disseminate only inexpensive techniques which can be purchased on producers' income and that only requires locally manufactured materials of common local use in villages (machetes, pickaxes, traditional axes, shovels, ladders).
- Disseminate techniques that are well mastered by the technical services of the environment or agriculture and which have already proved as effective.
- Examples of techniques ready to be disseminated include nurseries, plantations, direct sowing, loranthaceae parasitic control, rejuvenation cuts, grafting, Assisted Natural Regeneration (ANR) using easy to use systems of tracking (stakes or painted pebbles) and protection (branching, thorny bush placed on the seedlings...).

Suggested mode for dissemination of techniques

Dissemination of techniques should be done in two stages :

1. Training of local/endogenous trainers

- The identification of these trainers will be based on volunteering or selection by members of the group.
- Number of trainers to be trained: 5 to 10 per village
- Training will be provided by government environmental or agricultural extension technicians or research technicians who routinely practice these techniques in their professional activities
- They will be assisted by Serious Shea facilitators trained in the dissemination of techniques in local languages.

2. Cluster dissemination by local /endogenous trainers

- Other women producers in the village and surrounding villages will in turn be trained by endogenous trainers who have understood the techniques well. They will be assisted by Serious Shea facilitators and researchers.

6.7. Step 6. Participatory Monitoring and Evaluation

To be effective and motivate local people, the monitoring of training sessions and performance of techniques must be done by local people themselves under the supervision of environmental or agricultural officers and research technicians.

1. *Monitoring and evaluation of training sessions on an annual basis*

Examples of indicators

- Number of signed local resource management conventions
- Number of exchange trips organized for groups of women producers
- Number of study trip participants
- Number of endogenous trainers trained in the technique
- Number of producers trained in the technique
- Number of fields where the technique is applied
- Number of village nurseries set up
- Number of grafted shea wildlings
- Number of shea plants planted or transplanted
- Number of shea seeds sown
- Number of protected shea trees (RNA)
- Number of girdled shea trees
- Number of shea trees pruned against parasitic Loranthaceae
- Number of shea trees pruned for rejuvenation

2. *Monitoring and evaluating the performance of the techniques on a bi-annual basis at the start and end of the rainy season.*

Examples d'indicators :

- Number of plants produced in the nursery
- Number of successful transplants
- Number of shea plants planted or transplanted that have survived
- Number of shea plants resulting from seedlings
- Number of protected shea trees that have survived (RNA)
- Growth and development of protected shea plants compared to unprotected ones (increase in height, number of leaves)
- Number of girdled shea trees with increased production
- Number of shea plants pruned against the parasitic Loranthaceae having new leaves devoid of parasites
- Number of pruned shea trees for rejuvenation pruning that have resumed production

Implementation leaders

- Field owners
- Shea producers
- Forestry or Agriculture officers
- Research technicians
- Village Development Councils

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